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1. The conventional algorithm for evaluating a polynomial  $a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$  at  $x = c$  can be expressed in pseudocode by

```
1: procedure POLYNOMIAL( $c, a_0, a_1, \dots, a_n$  : real numbers)
2:    $power \leftarrow 1$ 
3:    $y \leftarrow a_0$ 
4:   for  $i = 1$  to  $n$  do
5:      $power \leftarrow power * c$ 
6:      $y \leftarrow y + a_i * power$ 
7:   end for
8:   return  $y$ 
9: end procedure
```

where the final value of  $y$  is the value of the polynomial at  $x = c$ .

- a) (10 points) Evaluate  $3x^2 + x + 1$  at  $x = 2$  by working through each step of the algorithm, showing the values assigned at each assignment step.
- b) (10 points) Exactly how many multiplications and additions are used to evaluate a polynomial of degree  $n$  at  $x = c$ ? (*Do not count additions used to increment the loop variable.*)
2. There is a more efficient algorithm (in terms of the number of multiplications and additions used) for evaluating polynomials than the conventional algorithm described in the previous exercise. It is called **Horner's method**. The following pseudocode shows how to use this method to find the value of  $a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$  at  $x = c$ .

```
1: procedure HORNER( $c, a_0, a_1, a_2, \dots, a_n$  : real numbers)
2:    $y \leftarrow a_n$ 
3:   for  $i = 1$  to  $n$  do
4:      $y \leftarrow y * c + a_{n-i}$ 
5:   end for
6:   return  $y$ 
7: end procedure
```

- a) (10 points) Evaluate  $3x^2 + x + 1$  at  $x = 2$  by working through each step of the algorithm, showing the values assigned at each assignment step.
- b) (10 points) Exactly how many multiplications and additions are used by this algorithm to evaluate a polynomial of degree  $n$  at  $x = c$ ? (*Do not count additions used to increment the loop variable.*)

3. Use the definition of “ $f$  is  $O(g(x))$ ” to prove that:
- (a) (5 points)  $x^4 + 9x^3 + 4x + 7$  is  $O(x^4)$ .
  - (b) (5 points)  $2^x + 17$  is  $O(3^x)$ .
  - (c) (5 points)  $\frac{x^2+1}{x+1}$  is  $O(x)$
  - (d) (5 points) for every positive integer  $k$ , it holds that  $1^k + 2^k + \dots + n^k$  is  $O(n^{k+1})$ .
4. Given a real number  $a$  and a nonnegative integer  $n$ , consider the following recursive method to compute  $a^n$ : Compute  $h = a^{\lfloor n/2 \rfloor}$  recursively. If  $n$  is even, return  $h \cdot h$ ; if  $n$  is odd, return  $a \cdot h \cdot h$ . When  $n = 0$ , return 1. Assume arithmetic operations take constant time.
- (a) (10 points) Find a worst-case running-time recurrence  $T(n)$  for this method.
  - (b) (10 points) Apply the Master Theorem to  $T(n)$  to obtain its time complexity.
5. Use generating functions to determine:
- (5 points) the number of different ways 10 identical balloons can be given to four children if each child receives at least two balloons.
  - (5 points) the number of different ways 12 identical action figures can be given to five children so that each child receives at most three action figures.
  - (5 points) the number of different ways 15 identical stuffed animals can be given to six children so that each child receives at least one but no more than three stuffed animals.
  - (5 points) the number of ways to choose a dozen bagels from three varieties—egg, salty, and plain—if at least two bagels of each kind but no more than three salty bagels are chosen.
6. Find the generating functions of each of the following sequences, in simple, closed form. In each case the sequence is defined for all  $n \geq 0$ .
- (a) (5 points)  $a_n = n$
  - (b) (5 points)  $a_n = 2n + 3$
  - (c) (5 points)  $a_n = n^2$
  - (d) (5 points)  $a_n = 2n^2 + 3n + 4$